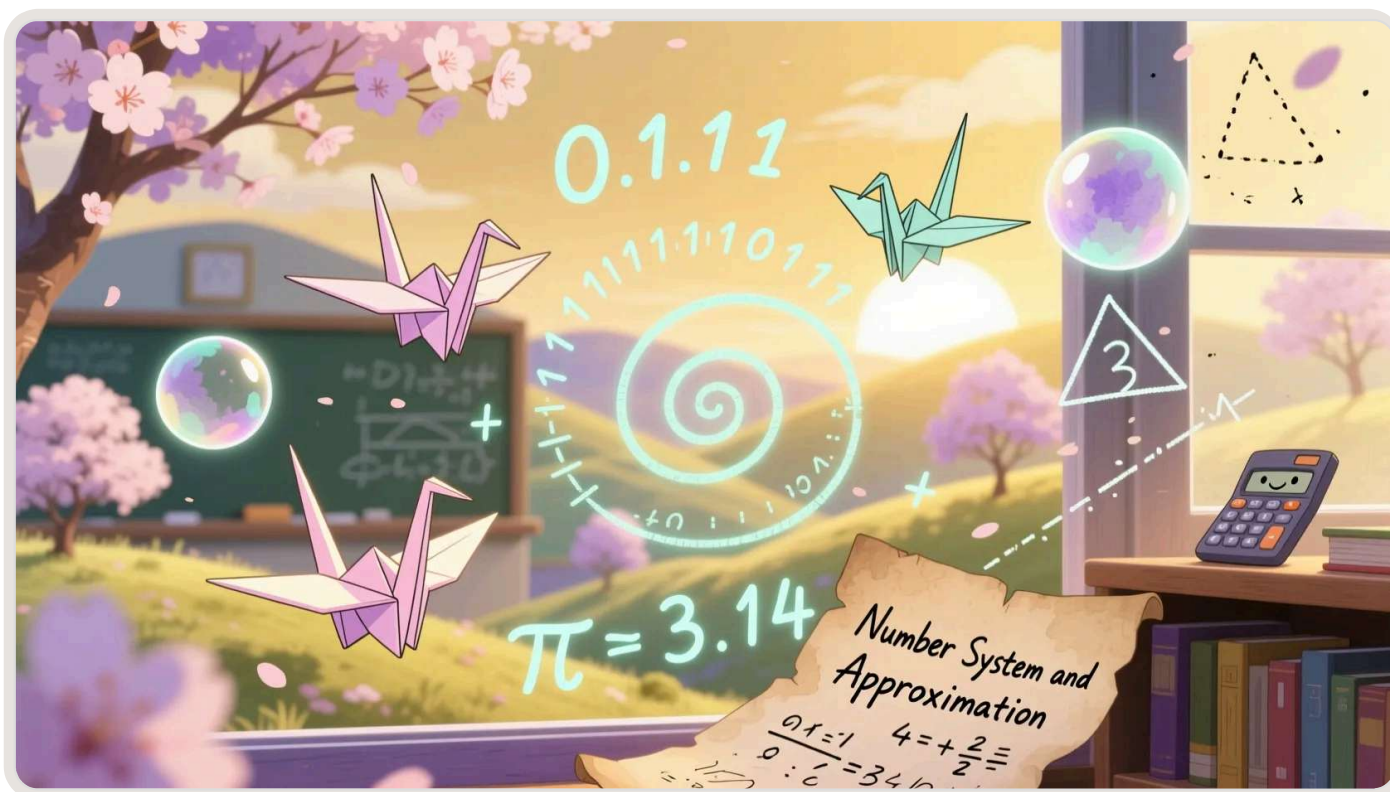




November 29, 2025 • Notes

Number System

This resource provides detailed notes about Number systems with Important points, examples and other things which are helpful for exams as well as contributes to better understanding of the topic.



The Number System is the framework used to represent and calculate quantities. Below are concise, exam-oriented notes starting from the basic building blocks up to Real numbers.

1. Hierarchy of Numbers

Understanding the “family tree” of numbers is the best way to remember them. Each subsequent type includes the previous ones (except for Irrational numbers, which are a separate branch).

- Natural Numbers ($\mathbb{N}$) $\subset$ Whole Numbers ($\mathbb{W}$) $\subset$ Integers ($\mathbb{Z}$) $\subset$ Rational Numbers ($\mathbb{Q}$)
- Real Numbers ($\mathbb{R}$) = Rational ($\mathbb{Q}$) + Irrational ($\mathbb{P}$)

2. Basic Types of Numbers

Natural Numbers ($\mathbb{N}$)

These are the basic “counting numbers” used in daily life.

- **Definition:** The set of positive integers starting from 1.
- **Range:** $1, 2, 3, 4, \dots \infty$
- **Key Exam Point:** The smallest natural number is **1**. There is no largest natural number.

Whole Numbers ($\mathbb{W}$)

These are Natural numbers plus zero.

- **Definition:** The set of natural numbers including **0**.
- **Range:** $0, 1, 2, 3, \dots \infty$
- **Key Exam Point:** Every Natural number is a Whole number, but **0** is a Whole number that is *not* a Natural number.

Integers ($\mathbb{Z}$)

This set expands to include negative values.

- **Definition:** Whole numbers plus their negative counterparts.
- **Range:** $\dots - 3, -2, -1, 0, 1, 2, 3 \dots$
- **Important:** 0 is neither positive nor negative.

3. Rational vs. Irrational Numbers (Crucial for Exams)

This distinction is often the source of most exam questions (MCQs and proofs).

Rational Numbers ($\mathbb{Q}$)

- **Definition:** Any number that can be written in the form $\frac{p}{q}$, where p and q are integers and $q \neq 0$.
- **Decimal Property:** Their decimal expansion is either **Terminating** (e.g., 0.5) OR **Non-terminating but Repeating** (e.g., 0.333...).
- **Examples:**
 - $\frac{1}{2}$
 - 5 (can be written as $\frac{5}{1}$)
 - 0 (can be written as $\frac{0}{1}$)
 - -3.5

Irrational Numbers

- **Definition:** Numbers that **cannot** be written as a simple fraction $\frac{p}{q}$.
- **Decimal Property:** Their decimal expansion is **Non-terminating AND Non-repeating**.
- **Examples:**
 - $\sqrt{2}, \sqrt{3}, \sqrt{5}$
 - π
 - $0.1010010001\dots$

Comparison Table: Rational vs. Irrational

Feature	Rational Numbers ($\mathbb{Q}$)	Irrational Numbers
Form	Can be written as $\frac{p}{q}$	Cannot be written as $\frac{p}{q}$
Decimal Type	Terminating or Repeating	Non-terminating and Non-repeating
Examples	$2, \frac{22}{7}, 0.333\dots, \sqrt{4}$	$\pi, \sqrt{2}, 1.414213\dots$
Perfect Squares	Square roots of perfect squares (e.g., $\sqrt{16} = 4$)	Square roots of non-perfect squares (e.g., $\sqrt{17}$)

4. Real Numbers and Exam Traps

Real Numbers ($\mathbb{R}$)

- **Definition:** The collection of **all** Rational and Irrational numbers together.
- **Visualization:** Every point on the number line represents a unique Real number.

Common Exam Traps and Notes

- **Is Zero Rational?** Yes, because $0 = \frac{0}{1}$.
- **Pi (π) vs. $\frac{22}{7}$**
 - π is **Irrational**.
 - $\frac{22}{7}$ is **Rational**.
 - $\frac{22}{7}$ is just an approximation of π .
- **Prime Roots:** The square root of any prime number ($\sqrt{2}, \sqrt{3}, \sqrt{5}$) is always irrational.
- **All Integers are Rational:** Any integer n can be written as $\frac{n}{1}$.

Divisibility and Divisibility Rules

Definition:

Divisibility means that when you divide one number by another, the result is a whole number with **zero remainder**.

- *Example:* 10 is divisible by 5 because $10 \div 5 = 2$ (remainder 0).
- *Counter Example:* 10 is not divisible by 3 because $10 \div 3 = 3.33\text{...}$ (remainder 1).

Divisibility Tests (1 to 20)

1. Basic Rules (Easy to remember)

Number	Rule	Example
1	All integers are divisible by 1.	5, 100, 987 are all divisible by 1.
2	Last digit is Even (0, 2, 4, 6, 8).	124 (ends in 4), 50 (ends in 0).
3	Sum of digits is divisible by 3.	123 $\rightarrow 1 + 2 + 3 = 6$. 6 is divisible by 3, so 123 is too.
5	Last digit is 0 or 5 .	145 (ends in 5), 290 (ends in 0).
6	Divisible by both 2 AND 3 .	24 : Even (divisible by 2) AND $2 + 4 = 6$ (divisible by 3).
9	Sum of digits is divisible by 9.	729 $\rightarrow 7 + 2 + 9 = 18$. 18 is divisible by 9.
10	Last digit is 0 .	540 ends in 0.

2. The “Last Digits” Family (Powers of 2)

Number	Rule	Example
4	Last 2 digits are divisible by 4.	312 : 12 is divisible by 4.
8	Last 3 digits are divisible by 8.	3816 : 816 is divisible by 8 ($816 \div 8 = 102$).
16	Last 4 digits are divisible by 16.	13264 : 3264 is divisible by 16 ($3264 \div 16 = 204$).

3. The “Chunking” Rule (For 7, 11, 13)

A powerful unified method for large numbers.

The Rule:

Form groups of **3 digits** from the right.

Subtract the sum of even-placed groups from the sum of odd-placed groups.

If the result is divisible by **7, 11, or 13**, the whole number is divisible.

For 7:

- *Standard Method:* Double the last digit and subtract from the rest.

- *Example:* $343 \rightarrow 34 - (3 \times 2) = 28$.
28 is divisible by 7.

For 11:

- *Standard Method:* Difference between the sum of digits in **odd** and **even** places must be 0 or a multiple of 11.
- *Example:* $1331 \rightarrow (1 + 3) - (3 + 1) = 4 - 4 = 0$.

For 13:

- *Standard Method:* Multiply last digit by 4 and add to the rest.
- *Example:* $169 \rightarrow 16 + (9 \times 4) = 16 + 36 = 52$.
52 is divisible by 13.

4. Composite Number Rules (Factor Methods)

Number	Components	Rule	Example: 420
12	3 and 4	Divisible by 3 AND 4 .	$4 + 2 + 0 = 6$ (div by 3). Ends in 20 (div by 4). Yes.
14	2 and 7	Divisible by 2 AND 7 .	Even. For 7: $42 - 0 = 42$. Yes.
15	3 and 5	Divisible by 3 AND 5 .	$4 + 2 + 0 = 6$ (div by 3). Ends in 0. Yes.
18	2 and 9	Divisible by 2 AND 9 .	Even. $4 + 2 + 0 = 6$ (not div by 9). No .
20	10 and 2	Ends in 0 AND tens digit even .	420 $\rightarrow$ ends in 0, tens digit 2 is even. Yes.

5. Special Prime Rules (17 and 19)

These use “osculation” techniques like 7 and 13.

osculation is a Vedic-maths method,

Simple idea (for divisibility)

For certain divisors (like 7, 13, 17, 19, etc.), you:

1. Take the **last digit** of the number.
2. Multiply it by a **fixed small number** (called the *osculator*).
3. **Add or subtract** this from the remaining part of the number.
4. Repeat until you get a small number that’s easy to check.

If that final small number is divisible by the divisor, the original big number is also divisible.

Example for 19 (osculator = 2, add):

- Check 361:
Last digit = 1 $\rightarrow 1 \times 2 = 2$
Remaining = 36 $\rightarrow 36 + 2 = 38$
38 is divisible by 19 $\rightarrow$ so 361 is divisible by 19.

17 (Negative Osculator 5)

Multiply last digit by 5 and subtract from the rest.

Example: 289

1. Last digit: 9 $\rightarrow 9 \times 5 = 45$
2. Subtract: $28 - 45 = -17$
3. 17 is divisible by 17 $\rightarrow \checkmark$

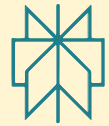
19 (Positive Osculator 2)

Multiply last digit by 2 and add to the rest.

Example: 361

1. Last digit: 1 $\rightarrow 1 \times 2 = 2$
2. Add: $36 + 2 = 38$
3. 38 is divisible by 19 $\rightarrow \checkmark$ (19×2)

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